

Arpit Sharma

CONTACT INFORMATION

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SUMMARY

Computer Scientist with 6+ years' experience designing and deploying scalable conversational AI and agentic systems. Proven innovator with patented contributions. Strategic leader skilled in mentoring teams and shaping corporate strategy to deliver AI-driven solutions for major retail use cases. Expertise in Large Language Models (LLMs), agentic frameworks, and AI workflow orchestration.

WORK EXPERIENCE

Walmart Associates Inc., Sunnyvale, CA, USA

Distinguished Data Scientist, June 2025 - Present

- Led the Intelligence team in the successful launch of multiple features within Walmart's Shopping Assistant, as well as several Minimum Viable Products (MVPs).
- Designed and implemented multi-agent workflows to support innovative shopping use cases, including event planning and essentials shopping, enhancing user experience and automation.
- Spearheading industry collaboration research initiatives focused on the development of low-cost, high-quality open-source LLM (Large Language Model) judges.

Principal Data Scientist, October 2022 - June 2025

- Provided technical leadership to a cross-functional team of over 15 engineers and scientists, guiding the multi-phase delivery of advanced machine learning systems.
- Designed and implemented an agentic evaluation framework for internal assessment of agentic workflows within a retail-focused chatbot.
- Defined quality and performance metrics for large language models (LLMs) across various retail use cases; additionally, developed an "LLM-as-a-judge" framework to evaluate these metrics.
- Led the fine-tuning of open-source LLMs to meet business-specific requirements for retrieval-augmented generation (RAG) and code-of-conduct compliance, utilizing techniques such as supervised fine-tuning (SFT) and direct preference optimization (DPO).
- Drove cross-organizational alignment by coordinating priorities, timelines, and success criteria across product, engineering, science, and operations teams—effectively resolving conflicting objectives and ensuring the timely delivery of high-impact solutions.
- Defined a schema for AI agents in retail environments and used it to develop multiple agent types, including RAG, Instruct, and Planning agents.
- Successfully led an academic research collaboration that resulted in a paper accepted at the international NLP conference NAACL 2025.

Staff Data Scientist, December 2020 - October 2022

- Developed and fine-tuned a BERT-based model for Natural Language Understanding (NLU) in a conversational AI application.
- Led the design and implementation of a clustering-based, self-serve intent prediction model for multi-domain chatbots, leveraging the Universal Sentence Encoder (USE) for text embeddings.
- Spearheaded a research initiative focused on domain-specific adaptation of DistilBERT for retail applications, incorporating universal dependency parsing techniques.

- Designed API contracts for the core intelligence module powering Walmart’s conversational AI platform.

Senior Data Scientist, September 2019 to December 2020

- Trained a Bi-LSTM + CRF model for intent classification and named entity recognition in Walmart’s shopping assistant.
- Researched and developed a deep learning-based intent classification model with contextual awareness for improved performance in multi-turn conversations.
- Designed and implemented a text normalization pipeline to preprocess and clean ill-formatted user inputs for Walmart’s shopping assistant, enhancing downstream model accuracy.

Arizona State University, Tempe, AZ, USA 85281 (May 2014 - December 2018)

- School of computing, informatics, & decision systems engineering
- Positions held: **Graduate Research Assistant and Graduate Teaching Assistant**
- Supervisor: *Dr. Chitta Baral, Dr. Joohyung Lee, Dr. Yu Zhang, Dr. Ryan Meuth, Dr. Mutsumi Nakamura, and Lect. Phillip Miller*

Walmart Associates Inc., Sunnyvale, CA, USA

Data Scientist Intern, May 2018 - August 2018

- Worked as a part of Voice eCommerce team. The responsibilities included research and development of a **semantic parser for Walmart eCommerce related customer queries**.

The Mathworks Inc., Natick, MA, USA

Application Support Engineer Intern, May 2013 - August 2013

- Worked as a part of Technical Support team. The responsibilities included **providing technical support to customers on MATLAB** and its toolboxes related issues. The major part of internship was spent on dealing with the issues related to Parallel Computing Toolbox from MATLAB. It included assignment of data to different processors, GPUs and the execution of same code on various processors parallelly.

Infosys Technologies Ltd., Mysore, INDIA

Systems Engineer, September 2010 - April 2012

- **Mainframe maintenance project (COBOL;JCL;DB2):** Worked on a life insurance project that was responsible for keeping track of distribution of commission to agents and was also responsible for keeping record of insurance policies of more than 13 million customers worldwide.

TECHNICAL SKILLS

Agentic AI, Conversational Systems & Machine Learning: Agentic Systems, Multi-agent orchestration, LLMs & NLP, LLM fine-tuning, Transformers, prompt engineering, Retrieval-Augmented Generation (RAG), RLHF, RLAI, PPO, DPO and GRPO for LLM alignment. Full ML life-cycle management. Expertise in advanced ML algorithms and deep learning frameworks (PyTorch, TensorFlow).

Core Technology & Architecture: Designing and building low-latency, fault-tolerant AI systems. Proficiency with GCP and Docker.

Ethical AI & Evaluation: Implementing safety, privacy, and compliance measures for conversational agents. Developing robust frameworks for evaluating and debugging complex agent behavior.

Programming: Java, Python, MATLAB, HTML, MongoDB, Oracle SQL, Neo4j.

EDUCATION

Arizona State University, Tempe, AZ, USA

Ph.D., Computer Science, January 2015 - August 2019

- Thesis Topic: *Towards Understanding Natural Language: Semantic Parsing, Commonsense Knowledge Acquisition and Applications*
- Advisor: Chitta Baral, Ph.D

M.S., Computer Science, August 2012 - December 2014

- Topic: *Solving Winograd Schema Challenge : Using Semantic Parsing, Automatic Knowledge Acquisition and Logical Reasoning*
- Advisor: Chitta Baral, Ph.D
- *Suma Cum Laude*

College Of Engineering Roorkee, Roorkee, INDIA

B.Tech., Computer Science, August 2006 - May 2010

- Grade: *75.2 on a scale of 100*
- *B.Tech with honors*

SELECTED PUBLICATIONS

1. Kun Qian, Maximillian Chen, Siyan Li, **Arpit Sharma**, Zhou Yu. “Bottom-Up Synthesis of Knowledge-Grounded Task-Oriented Dialogues with Iteratively Self-Refined Prompts”. Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL) 2025 (Accepted in Main Conference)
2. Pratik Jayarao and **Arpit Sharma**. “Retraining DistilBERT for a Voice Shopping Assistant by Using Universal Dependencies” Proceedings of The Fourth Workshop on Reasoning and Learning for Human-Machine Dialogues. 2021.
3. **Arpit Sharma**. “Improving Intent Classification in an E-commerce Voice Assistant by Using Inter-Utterance Context.” Proceedings of The 3rd Workshop on e-Commerce and NLP. 2020.
4. Ashok Prakash, **Arpit Sharma**, Arindam Mitra and Chitta Baral. “Combining Knowledge Hunting and Neural Language Models to Solve the Winograd Schema Challenge” Proceedings of the 57th Conference of the Association for Computational Linguistics, 2019.
5. **Arpit Sharma**. “Using Answer Set Programming for Commonsense Reasoning in the Winograd Schema Challenge” Theory and Practice of Logic Programming 19 (5-6). Cambridge University Press: 1021–37. doi:10.1017/S1471068419000334, 2019.
6. **Arpit Sharma**, Nguyen H. Vo, Somak Aditya, and Chitta Baral. “Towards Addressing the Winograd Schema Challenge-Building and Using a Semantic Parser and a Knowledge Hunting Module.” In International Joint Conference on Artificial Intelligence, 2015.

SELECTED
PRESENTATIONS

The 3rd Workshop on e-Commerce and NLP (ECNLP). 2020. Held Virtually with ACL 2020 July 2020

- Improving Intent Classification in an E-commerce Voice Assistant by Using Inter-Utterance Context.
- Presentation available on [Slideshare](#)

35th International Conference on Logic Programming (ICLP) 2019, Las Cruces, New Mexico, USA September 2019

- Using Answer Set Programming for Commonsense Reasoning in the Winograd Schema Challenge
- Presentation available on [Youtube](#)

SELECTED
PATENTS

1. Jayarao, Pratik Sridatt, **Arpit Sharma**, and Deepa Mohan. "Methods and apparatus for natural language understanding in conversational systems using machine learning processes." U.S. Patent 11,960,842, issued April 16, 2024. (Published 2024)
2. **Arpit Sharma**, Vivek Kaul, and Shankara Bhargava Subramanya. "Method and apparatus for parsing and representation of digital inquiry related natural language." U.S. Patent No. 11,568,007. 31 Jan. 2023. (Published 2023)
3. Hafshejani, Ghodratollah Aalipour, and **Arpit Sharma**. "Methods and apparatus for retrieving information via an intermediate representation." U.S. Patent No. 11,650,980. 16 May 2023. (Published 2023)
4. **Arpit Sharma**. "Methods and apparatus for improving search retrieval using inter-utterance context." U.S. Patent No. 11,715,469. 1 Aug. 2023. (Published 2023)
5. Hafshejani, Ghodratollah Aalipour, **Arpit Sharma**, Sahana Radhakrishna, C. H. E. N. Haoxuan, Tanmay Peter Kuruvilla, Phani Ram Sayapaneni, and Kotlasanthos Kumar. "Methods and apparatus for refining a search." U.S. Patent Application 17/163,087. (Published 2023)
6. **Arpit Sharma**. "System and Method for Determining a Real-Time Response Based on an Understanding of the Conversational Context." U.S. Patent Application 63/441,531. (Applied 2023)

PEER REVIEWER

1. Official reviewer for NORA'25 - Workshop on KNOWledge GRaphs & Agentic Systems Interplay. Collocated with NeurIPS'25
2. Official reviewer for Empirical Methods in Natural Language Processing (EMNLP) Industry Track 2024.
3. Official member of the program committee for The 29th International Conference on Computational Linguistics in 2022 (COLING 2022).
4. Official reviewer of the Association for Computational Linguistics conferences since October 2021 (ARR Oct 2021 - present)
5. Official member of the program committee for Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2021)
6. Official NLP Applications track reviewer for The 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020
7. Official member of the program committee for The 28th International Conference on Computational Linguistics (COLING 2020) Industry track

AWARDS

- **Graduate College Completion Fellowship** award for Spring 2019.
- **University Graduate Fellowship** award for 2018, 2017 and 2016.
- Computer Science Department award for **Outstanding Master's Student** 2014-15